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Object-free backdoor attack and defense on semantic segmentation

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ABSTRACT

Deep neural networks (DNNs) have proven to be applied in various fields, such as autonomous driving. However, it is compromised to a severe threat, namely a backdoor attack, which maliciously manipulates a model by poisoning a small portion of training data. Previous works mainly focused on backdoor attacks in image classification, however, it is scarcely studied in semantic segmentation widely used in computer vision. In this paper, we proposed a novel backdoor attack on semantic segmentation named object-free backdoor attack (OFBA). This method allows the free selection of object classes to be attacked during the inference, breaking the limitation of previous work only attacking object classes determined before training. In addition, to identify the security of unknown models and mitigate the threat of OFBA, we proposed a defense method called Segment Cleanse, alleviating the dilemma of insufficient computing resources and high detection error rates in Neural Cleanse. The experimental results show that our proposed methods are always effective on state-of-the-art models.

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1. Introduction

Computer vision has been critical in various application scenarios over the last decade. Due to the extraordinary performance of DNNs (ResNet (He et al., 2016a), VGG (Simonyan and Zisserman, 2015), DenseNet (Huang et al., 2017), etc.), they are widely used in most computer vision tasks, such as image classification (He et al., 2016b; Tan and Le, 2019), image segmentation (Hong et al., 2023; Long et al., 2015), and object detection (Li et al., 2022b; Zhang et al., 2022). However, while enjoying the conveniences of DNNs, humans also bear huge risks caused by their inexplicability, among which backdoor attack is one of the most well-known risks (Li et al., 2021b; Wang et al., 2022a; Xue et al., 2022). As such, these risks hinder development in some critical areas, such as automatic driving (Janai et al., 2020) and healthcare (Esteva et al., 2017).

Recently, due to the complexity of practical tasks, training DNNs requires a terrific amount of data and computational resources. To meet these demands, resources, such as databases, servers, and even models sourced from third-party platforms, introduce opacity during the training, opening the door to malicious backdoor attackers.

Backdoor attacks have recently gained attention in numerous fields, especially image classification. A successful backdoor attack in the classification task allows attackers to inject a trigger into several images, modify their labels to the target label, and then mix these poisoned examples into the training set. During the inference, the infected model will behave maliciously on poisoned images injected with the trigger while working normally on clean images (Gu et al., 2019). To our knowledge, the backdoor attack and defense have yet to be extensively studied outside standard classification tasks. Therefore, the backdoor robustness of DNNs in complex scenarios, especially in semantic segmentation, still needs to be further explored. Fig. 1 illustrates the process of semantic segmentation, which is obviously much more complex than image classification. This paper focuses on semantic segmentation's backdoor attack and defense, which is more challenging than image classification.

The backdoor attack on classification (Chen et al., 2017b; Gu et al., 2019), often assigns the same target label for all poisoned images, which is irrelevant to their ground truths. A previous work achieves the first object-level backdoor attack called fine-grained backdoor attack (FGBA) (Li et al., 2021a), in which only pixels of objects with the attacked class (*i.e.*, the attacker-specified class for

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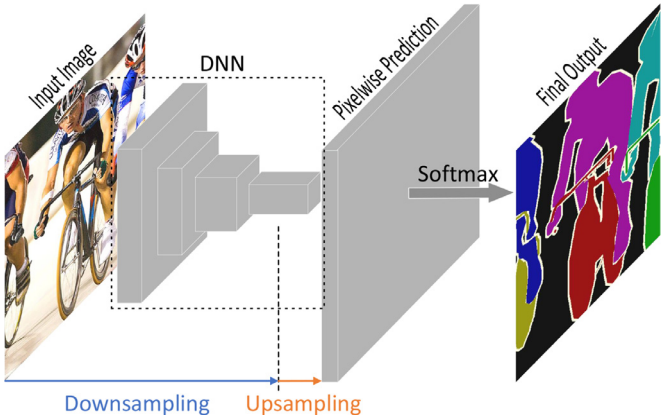

Fig. 1. Illustration of a semantic segmentation process. The input image is downsampled with features extracted by DNN (ResNet, VGG, DenseNet, etc), then these features are upsampled as pixelwise prediction. Finally the pixelwise prediction results are presented by different object classes..

misclassifying) will be labeled with the target class. That is to say, instead of associating a trigger with a single label (*i.e.*, the target label), the infected model establishes a strong association between the trigger and an object class (*i.e.*, the target class). It also means that the attack is already at the object level, not just at the image level.

We proposed a novel backdoor attack, namely an object-free backdoor attack (OFBA). It can select several objects within an image to manipulate their predictions and keep the remaining objects still able to be segmented to their ground truths during the inference. Instead of setting the fixed attacked class before training in FGBA (Li et al., 2021a), we continuously change the attacked class and the trigger position during the training. As a result, the same image will be assigned different target labels when the trigger appears in various positions. During the inference, in contrast to the fixed attacked class in FGBA, the attacked class in our attack is variable. That is why we only attack one model to selectively manipulate the prediction of any object. An example of target labels generated by different backdoor attacks is shown in Fig. 2. Experiments verify that our method can achieve almost the same attack success rate as previous works on three state-of-the-art models, including DeepLabv3 (Chen et al., 2017a), DenseASPP (Yang et al., 2018), and DaNet (Fu et al., 2019). Meanwhile, the damage caused by our attack on applying semantic segmentation models is more than negligible.

Since datasets and models on semantic segmentation are more complex than those on classification, defense for classification tasks cannot fully apply to segmentation tasks. To protect semantic segmentation models from backdoor attacks, such as OFBA, we analyze the limitations of Neural Cleanse (Wang et al., 2019), a mainstream defense on classification, which can be summarised as the infiniteness of potential target labels and the dispersion of background in the object class distribution space. Then, we propose Segmentation Cleanse, which extends Neural Cleanse to the object level and combines the ideas of bootstrap aggregating (Breiman, 1996) during anomaly detection. Numerous experiments show that the proposed defense can accurately identify whether the model is infected and significantly mitigate the infected model to bring it back to normal after retraining.

The main contributions of our work are summarized as follows:

- We propose a novel backdoor attack (OFBA) to achieve more severe malicious manipulation of semantic segmentation. The attackers can arbitrarily select object classes to attack during the inference without pre-determining the attacked class before training.
- We explore various components' effects on OFBA, including residual structures, skip-connections, and dilated convolutions. We find that most components contributing to the performance of models can encourage the creation of a backdoor.
- We analyze the non-applicability of Neural Cleanse against OFBA. Based on the analysis, we propose a method called Segment Cleanse. It can detect whether the semantic segmentation model is anomalous and mitigate the anomalous model to make it work properly.
- Evaluation metrics for backdoor attacks on semantic segmentation are presented. We conduct extensive experiments to verify the effectiveness of our attack and defense.

2. Related work

In this section, we mainly introduce related works covered by this paper, including semantic segmentation, backdoor attack, and the existing defense strategy. Table 1 summarizes the existing backdoor attacks and defense strategies mentioned in this section, including the capacity of attackers or defenders in these papers.

2.1. Semantic segmentation

As the pioneer of semantic segmentation in deep learning, FCN (Long et al., 2015) decomposed the model into two significant parts encoder and decoder. Olaf *et al.* (Ronneberger et al., 2015) adopted a skip-connection to stitch the low-level feature information acquired by the encoder and the high-level feature information obtained by the decoder, thus compensating for the loss of image edge detail information due to fast-down sampling. Chen *et al.* (Chen et al., 2018) proposed the DeepLab series, which increased the perceptual wildness of the models by adding components such as dilated convolution, ASPP, etc. Yang et al. (2018) used the DenseNet structure to densely connect multi-sized ASPPs. DaNet (Fu et al., 2019), on the other hand, was a traditional FCN with two types of attention modules attached to it, modeling semantic interdependencies in the spatial and channel dimensions, respectively.

So far, research on the accuracy of models on semantic segmentation has reached a more mature stage, while the security of these models is still relatively little explored. Previously, many studies (Arnab et al., 2020; Gu et al., 2022; Metzen et al., 2017; Xie et al., 2017) have shown the vulnerability of semantic segmentation models to adversarial attacks, mainly using gradient ascent to find some small perturbations so that the corresponding image cannot be segmented by the models (*e.g.,* the output of the model will contain various objects that are not present in the image). Unlike adversarial attacks, backdoor attacks only require the training dataset to be poisoned without any information about the model. This paper is concerned with the vulnerability that semantic segmentation models exhibit in the face of backdoor attacks.

2.2. Backdoor attack

The first work (Gu et al., 2019) of backdoor attacks was in the field of image classification, which poison partial training data by attaching a trigger to the bottom right corner of the images and modifying their labels to cause target misclassification. Chen *et al.* (Chen et al., 2017b) proposed a black-box attack simultaneously, where the user does not know the model and training examples. They poison examples through blend injection, and its trigger is relatively undetectable compared to BadNets. In subsequent work, Salem et al. (2022) studied the sample-specific backdoor attack by controlling the loss function during the training. Li *et al.* Li et al. (2021b) avoided control over the training process

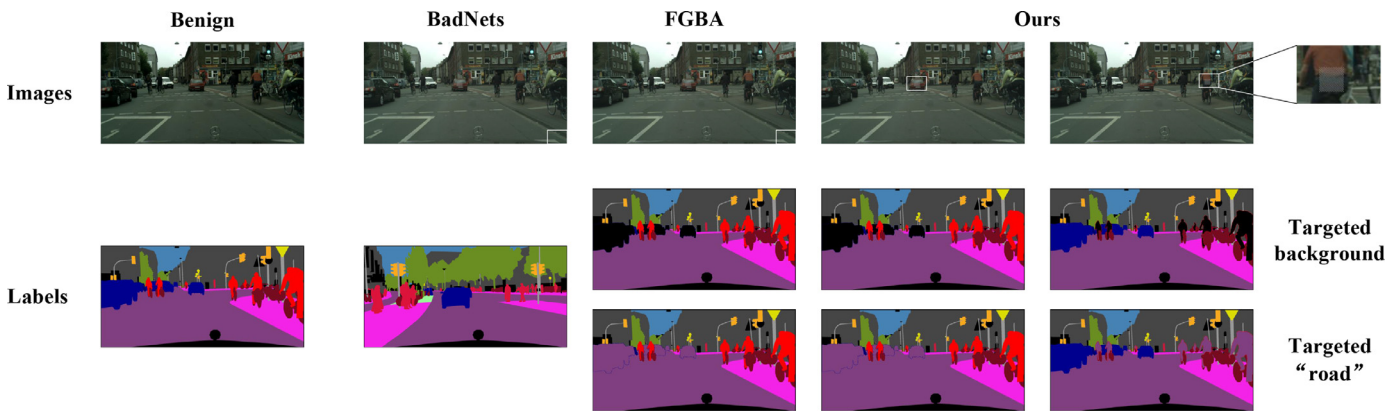

Fig. 2. Comparison between our proposed OFBA and previous attacks (BadNets and FGBA). The first row shows all the images, both clean and poisoned, where the white box indicates the position of the trigger. The following two rows show the ground truth and the target labels generated by each attack method. In the object-level attacks (FGBA and our attack), we display the target labels when background (black areas in labels) and "road" (purple areas in labels) become the target class in the second and third rows, respectively. In this example, BadNets has a constant target label irrelevant to the image. In the case of FGBA, pixels of objects with the attacked class "car" (blue areas in the ground truth) are labeled with the target class. For our target labels, when the trigger is located in a car, all pixels of objects with "car" are labeled with the target class. When the trigger is located in a person, pixels of objects with "person" (red areas in the ground truth) are labeled with the target class.

Table 1
A summary of existing backdoor attacks and defense strategies mentioned in Related Work. We list previous literature on backdoor attacks and defenses in different scenarios and the attacker or defender capability they assume, including whether a partial or complete training set is required and whether the training schedule needs to be controlled. Chan et al. (2022) and our work design both the backdoor attack and defense in their papers, and we illustrate the capacities of the attacker and defender, respectively.

Literature	Year	Scenarios	Type		Attacker or Defender Capacity	
			Attack	Defense	Training Set	Training Schedule
Chen et al. (2017b)	2017	classification	√		√	
Liu et al. (2018)	2018	classification		√	√	√
Gu et al. (2019)	2019	classification	√		√	
Gao et al. (2019)	2019	classification		√	√	
Wang et al. (2019)	2019	classification		√	√	√
Qiao et al. (2019)	2019	classification		√	√	√
Zhu et al. (2020)	2020	classification		√	√	√
Li et al. (2021b)	2021	classification	√		√	
Zhai et al. (2021)	2021	speech recognition	√		√	
Li et al. (2021a)	2021	semantic segmentation	√		√	
Wu and Wang (2021)	2021	classification		√	√	√
Qi et al. (2021)	2021	NLP		√	√	
Shao et al. (2021)	2021	NLP		√	√	√
Fan et al. (2021)	2021	NLP		√	√	
Salem et al. (2022)	2022	classification	√		√	√
Zhao et al. (2022)	2022	classification	√		√	√
Wang et al. (2022b)	2022	classification	√		√	√
Hong et al. (2022)	2022	classification	√			√
Zhong et al. (2022)	2022	classification	√		√	√
Bagdasaryan and Shmatikov (2022)	2022	NLP	√		√	√
Shao et al. (2022)	2022	NLP	√		√	
Liu et al. (2022a)	2022	NLP	√		√	
Gan et al. (2022)	2022	NLP	√		√	
Chan et al. (2022)	2022	object detection	√		√	
			√	√	√	
Li et al. (2022c)	2022	object detection	√		√	
Ma et al. (2022)	2022	object detection	√		√	√
Wu et al. (2022)	2022	object detection	√		√	
Li et al. (2022a)	2022	image captioning	√		√	
Ye et al. (2022)	2022	speech recognition	√		√	
Ze et al. (2022)	2022	speech recognition	√		√	
Feng et al. (2022)	2022	classification object detection semantic segmentation	√		√	
Guan et al. (2022)	2022	classification		√	√	√
Liu et al. (2022b)	2022	classification		√	√	√
Chen et al. (2022)	2022	classification		√	√	√
Ours		semantic segmentation	√		√	
				√	√	√

and thus presented a more covert sample-specific attack. More recently, unlike previous works that only considered the effectiveness and stealth of attacks, more and more works (Hong et al., 2022; Wang et al., 2022b; Zhao et al., 2022; Zhong et al., 2022) have focused on how to allow backdoor attacks to bypass existing backdoor defense strategies.

Based on previous works focusing on image classification, a series of backdoor attacks focused on various specific scenarios have been proposed, such as natural language processing (Bagdasaryan and Shmatikov, 2022; Gan et al., 2022; Liu et al., 2022a; Shao et al., 2022), object detection (Chan et al., 2022; Li et al., 2022c; Ma et al., 2022; Wu et al., 2022), image captioning (Li et al., 2022a), speech recognition (Ye et al., 2022; Ze et al., 2022; Zhai et al., 2021), etc. In particular, in semantic segmentation, Li et al. (2021a) verified experimentally that a backdoor attack is also a massive threat in semantic segmentation models and proposed an object-level backdoor attack that is different from the attack on classification. In the same year, Feng et al. (2022) achieved a successful backdoor attack on medical oncology segmentation, which is similar to BadNets described in Fig. 2. However, the previous works required setting up the attacked classes in advance, which led to the limitation of deploying backdoor attacks in real-world scenarios (*i.e.,* only one or a few fixed object classes could be attacked during the inference). Hence, we establish a strong association between the trigger and all object classes, which enables us to freely select object classes to attack during the inference.

2.3. Existing defense strategy

Current research on backdoor defense strategies focuses on image classification, which can be broadly categorized as preprocessing (Gao et al., 2019), model reconstruction (Liu et al., 2018; Wu and Wang, 2021), trigger synthesis (Guan et al., 2022), and model diagnosis (Liu et al., 2022b). Among them, trigger synthesis had an excellent performance as one of the mainstream strategies. Wang et al. (2019) proposed the first defense based on trigger synthesis in the white-box scenario, *i.e.*, Neural Cleanse. They reversed engineering to recover the potential triggers by solving an optimization problem. Qiao et al. (2019) found that the trigger recovered by Neural Cleanse was usually significantly different from the original trigger and discussed the generalizability of the backdoor trigger. Similar ideas were working in recent works (Chen et al., 2022; Guan et al., 2022; Zhu et al., 2020), where they proposed better objective functions for reverse engineering.

In other application scenarios, there are a small number of backdoor defense strategies (Fan et al., 2021; Qi et al., 2021; Shao et al., 2021) on NLP. Besides, Chan et al. (2022) explored the feasibility of defending against backdoor attacks on object detection. However, to our knowledge, no defense strategy against backdoor attacks on semantic segmentation has been proposed. When using Nueral Cleanse (Wang et al., 2019) directly to defend against backdoor attacks on semantic segmentation, we will face huge computational consumption and a high detection error rate. To this end, we analyze the reason and propose a defense strategy that is specifically applicable to semantic segmentation models.

3. Background

3.1. Notations of semantic segmentation

A semantic segmentation model aims to segment the objects within an image, which outputs a 2D metric of the same size (*i.e.*, length and width) as the image. Let $\mathcal{D} = (\boldsymbol{x}_i, \boldsymbol{y}_i)_{i=0}^{N-1}$ denotes a dataset containing N examples and K object classes. $\boldsymbol{x}_i \in \{0, 1, \ldots, 255\}^{C\times W\times H}$ is an image, $\boldsymbol{y}_i \in \{0, 1, \ldots, K-1\}^{W\times H}$ is the label of the image $\boldsymbol{x}_i$, where C, W, and H respectively denote the numbers of channel, the width, and height of the image $\boldsymbol{x}_i$ (*e.g.,* C of a 2D color image is usually 3). Consider a standard supervised semantic segmentation task, we hope to learn a mapping function $\mathcal{F}_\theta : \mathcal{X} \to \mathcal{Y}$, where $\mathcal{X}$ is the input domain, and $\mathcal{Y}$ is the set of labels composed of object classes. The task is to optimize the parameters θ by minimizing target function $1/N \sum_i \mathcal{L}(\mathcal{F}_\theta(\boldsymbol{x}_i), \boldsymbol{y}_i)$, where $\mathcal{L}(\cdot)$ denotes a loss function, *e.g.*, cross entropy. Essentially, $\mathcal{F}_\theta$ aims to classify each pixel to the correct object class with a high confidence score. For an object class, the ratio of the intersection and union between prediction and ground truth is defined as intersection-over-union (*IoU*). *mIoU* represents the mean of *IoU*s for all object classes. The proportion of correctly classified pixels to all pixels is defined as pixel accuracy (*mPA*). *mPA* and *mIoU* are the most common evaluation metrics for segmentation tasks.

3.2. General pipeline of backdoor attacks

Following the standard training scheme of backdoor attacks (Chen et al., 2017b; Gu et al., 2019; Li et al., 2021b), an infected model is trained with the backdoor dataset $\mathcal{D}'$, which combines the benign and poisoned examples. To create $\mathcal{D}'$, randomly select $\gamma|\mathcal{D}|$ examples from $\mathcal{D}$, where γ is a poisoning rate, and modify them by some specific means. In this way, a benign example $(\boldsymbol{x}, \boldsymbol{y})$ is transformed into a poisoned example $(B(\boldsymbol{x}), A(\boldsymbol{y}))$, where $B(\cdot)$ and $A(\cdot)$ denote trigger-injection function and label-conversion function, respectively. After training a model $\mathcal{F}_\theta$ with $\mathcal{D}'$, we alter the behavior of $\mathcal{F}_\theta$ to satisfy Equation (1), (2).

$$\mathcal{F}_\theta(B(\boldsymbol{x})) = A(\boldsymbol{y}) \tag{1}$$

$$\mathcal{F}_\theta(\boldsymbol{x}) = \boldsymbol{y} \tag{2}$$

In this paper, we focuses on constructing $B(\cdot)$ and $A(\cdot)$, making them applicable to segmentation tasks. Besides, we defend against the proposed attack by improving the defense for classification models. We advance the state-of-the-art by further enhancing the utility of the backdoor attack and defense in complex domains.

4. Methodology

In this section, we will describe the details of our proposed attack and defense methods. Section 4.1 introduces the threat model, which includes implementation scenarios and task goals. The threat model can be divided into two parties, an *attacker*, who wishes to manipulate a DNN, and a *defender*, who wishes to protect a DNN from the backdoor attack. Section 4.2 describes how to produce several effective poisoned examples of OFBA. First, we design a trigger-injection function. Through this function, the trigger is injected into several objects within an image by Chen et al. (2017b), where the areas of pixels changed by the same trigger all belong to the same object. Then, we modify the label of the poisoned image by the label-conversion function, where only pixels of objects comprising the trigger are labeled with the target class. In Section 4.3, we find that the main difference between the semantic segmentation dataset and classification dataset is their labels and input spaces. Based on this discovery, we propose Segment Cleanse to detect and mitigate models attacked by OFBA. Fig. 3 describes the pipeline of our works mentioned above.

4.1. Threat model

From the attacker's perspective, we assume the weakest attack scenario, where only the training set is accessible and can be modified. At the same time, they cannot access or even control other parts, such as the loss function, during the training and have no information about the inference. This is a typical setting for backdoor

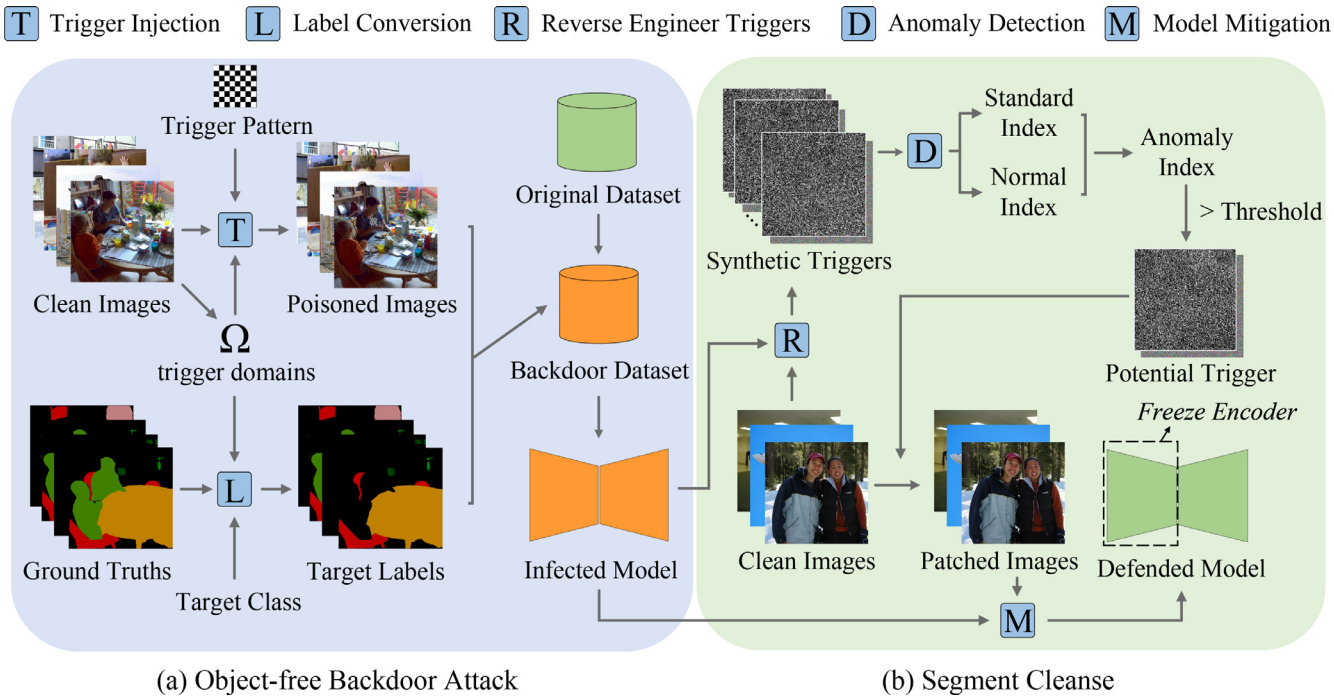

Fig. 3. The Overview framework of the proposed attack (OFBA) and defense (Segment Cleanse). The generation of poisoned examples based on OFBA is shown in (a). The process of detecting and mitigating models based on Segment Cleanse is shown in (b). In the above framework, the blue boxes represent important processes involved in carrying out the attack or defense implementation, which are focused on in this paper. Specifically, "Trigger Injection" and "Label Conversion" are done via two functions, *i.e.,* Equation (3) and Equation (4), both of which depend on the trigger domain we have defined. "Reverse Engineer Triggers" is an optimization problem that is defined as Equation (7). "Anomaly Detection" can be defined as Equation (9), (10) and "Model Mitigation" is to retrain the infected model with patched images.

attackers (Chen et al., 2017b; Gu et al., 2019), which allows the attacker to silently carry out malicious attacks in various real-world scenarios, such as adopting third-party data sources. As for the defender, we follow previous works (Wang et al., 2019), in which the defender can get a small number of clean images from the training set and obtain complete control of the infected model.

In a successful object-free backdoor attack, the *attacker* can select the pixels of objects within poisoned images to be classified to the target class. Meanwhile, the infected model can achieve comparable segmentation performance to the benign model for other objects within poisoned images and all objects within clean images. In addition, the *defender* aims to identify whether an unknown model is anomalous and which object class may be the target. Then, the defender can mitigate the anomalous model to work normally on all images, including poisoned and clean images.

4.2. Object-free backdoor attack (OFBA)

In this section, we illustrate our proposed object-free backdoor attack. Generating a batch of poisoned examples is generally divided into two processes: trigger injection and label conversion. In general, the trigger injection process injects the predefined pattern or even feature into clean images in a specific way. The label conversion process converts labels of poisoned images into target labels according to the attacker's intention, thus using what appears to be the wrong label to human eyes for the model's recognition.

Specifically, we first need to find a location to inject the trigger, where all pixels belong to the same objects. Then, we convert these object classes to the target class. Therefore, we present the following definition.

Definition 1 (. *trigger domain*) For an input instance $\boldsymbol{x}$ with size $C \times W \times H$ and a trigger pattern Δ with size $C \times w \times h$, we assume the input instance $\boldsymbol{x}$ and the trigger pattern Δ are both in their vector representation. let $P(i, j)$ represents the object class to which the pixel at position (i, j) in the input instance $\boldsymbol{x}$ belongs. (i_0, j_0) is a position in matrix x, where $0 \le i_0 \le W - w$, $0 \le j_0 \le H - h$. Meanwhile, it satisfies $P(i_x, j_x) = P(i_0, j_0)$ for any position (i_x, j_x), where $i_0 \le i_x \le i_0 + w$ and $j_0 \le j_x \le j_0 + h$. (i_0, j_0) is called a trigger point, and the positions set Ω_x consisting of all positions (i_x, j_x) for (i_0, j_0) is called a trigger domain. Fig. 4 illustrates a qualified trigger point and its trigger domain.

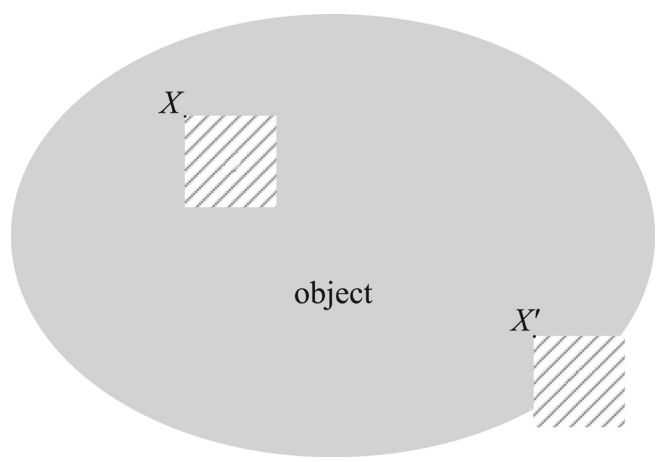

Fig. 4. A diagram of a trigger point and its trigger domain. A grey oval is an object in the image in which all the pixels belong to the same object class. (X, Ω) is a combination of a compliant trigger point and its trigger domain, which meets the requirements in Definition 1. In contrast, X' cannot be used as an ideal trigger point for this object, which is a conclusion that the object does not fully contain Ω'.

Remark 1. In this paper, we choose the black and white checkboard as the trigger pattern. When the trigger pattern is non-rectangular, we can use the tangent matrix of the pattern as the

trigger matrix, and the excess of non-patterned positions inside the rectangle should be filled with $[0, 0, \ldots, 0]^C$.

4.2.1. Trigger injection

For an input instance $\boldsymbol{x}$ from a semantic segmentation dataset $\mathcal{D}$ containing K object classes, suppose the attacker poisons m different object classes. $\left(i_0{}^0, j_0{}^0\right), \left(i_0{}^1, j_0{}^1\right), \ldots, \left(i_0{}^{m-1}, j_0{}^{m-1}\right)$ denote m trigger points from different objects, and their corresponding trigger domains are represented as $\Omega_x{}^0, \Omega_x{}^1, \ldots, \Omega_x{}^{m-1}$. We mix the pixels in each trigger domain with the trigger pattern Δ through the Blend (Chen et al., 2017b) technology to generate a poisoned image. Assume the input instance $\boldsymbol{x}$ and the trigger pattern Δ are both in their vector representations, the object-free trigger-injection function $B^{OFBA}(\cdot)$ is defined as follows:

$$B^{OFBA}(\boldsymbol{x}, \lambda, \Delta)_{c,i,j} = \begin{cases} (1-\lambda)\cdot \boldsymbol{x}_{c,i,j} + \lambda \cdot \Delta_{c,i',j'} & if\ (i,j) \in \Omega_x{}^k \\ \boldsymbol{x}_{c,i,j} & others \end{cases} \tag{3}$$

where $k \in \{0, 1, \ldots, m-1\}$, $i' = i - i_0{}^k$, $j' = j - j_0{}^k$. $\boldsymbol{x}_{c,i,j}$ and $\Delta_{c,i,j}$ denote two values corresponding to the position (c, i, j) in the input instance $\boldsymbol{x}$ and the trigger pattern Δ, respectively. $\lambda \in [0, 1]$ is a visible coefficient, controlling the extent to which triggers cover pixels in the trigger domains. When $\lambda = 1$, pixels of the trigger completely overwrite the original pixels, and when $\lambda = 0$, the original pixels value is not modified. It can be seen that the smaller the visible coefficient, the easier it is for the human eye to ignore the presence of the trigger pattern, but the more difficult it is for the model to learn this feature transformation. Meanwhile, $P\left(i_0{}^0, j_0{}^0\right), P\left(i_0{}^1, j_0{}^1\right), \ldots, P\left(i_0{}^{m-1}, j_0{}^{m-1}\right)$ are attacked classes we have chosen in this image, which can be variable in different images. When the number of poisoned samples is much larger than the number of object classes, only the attacked classes need to be chosen randomly for each poisoned image during the training. At this moment, the set of attacked classes in all poisoned images will most likely contain all object classes, which is a prerequisite for being able to attack any object class during inference.

4.2.2. Label conversion

The label-conversion function often controls the process of label conversion. Accordingly, we propose the object-free label-conversion function $A^{OFBA}(\cdot)$ to generate target labels of poisoned images, which can be defined as follows:

$$A^{OFBA}(\boldsymbol{x}, tar)_{i,j} = \begin{cases} tar & if\ P(i,j) = P\left(i_0{}^k, j_0{}^k\right) \\ P(i,j) & others \end{cases} \tag{4}$$

where $k \in \{0, 1, \ldots, m-1\}$. tar denotes the target class specified by the attacker. Note that the trigger domain set will be empty if the size of the trigger pattern is too large to find a qualifying trigger point. At this moment, this poisoned example is no different from the benign example. This is one of the reasons why it failed to attack some objects in some images during the inference. On the contrary, when the size of the trigger pattern is too small, the trigger may be too inconspicuous to be used as a significant feature for model recognition. It will result in a low attack success rate or even failure to train an infected model. Therefore, choosing a suitable trigger size is crucial to achieving a successful backdoor attack.

4.3. Segmentation cleanse

Since previous methods cannot be generalized to OFBA, we propose Segmentation Cleanse. It is improved on Neural Cleanse (Wang et al., 2019), which assumes the defender can only access very few but complete (containing all the object classes) clean images. Segmentation Cleanse can detect and mitigate infected models attacked by OFBA. In this section, we describe the details of our technique to defend OFBA.

4.3.1. Reverse engineer triggers

Defenses based on trigger synthesis (Guan et al., 2022; Wang et al., 2019; Zhu et al., 2020) are mainstream backdoor defense methods for image classification. They obtain synthetic triggers towards every potential target label through neuron reverse engineering or generative modeling and determine whether the model is infected based on an anomaly detector. However, a single label in the semantic segmentation dataset is a collection of multiple object classes, not a single value. Given a semantic segmentation dataset containing K object classes and consisting of images with size $C \times h \times w$, there exist K^{hw} potential target labels, that is to say, the number of synthetic triggers to be reversed is K^{hw}. In this situation, the amount of computation required is currently unaffordable. Therefore, we generalize the reverse engineering to the object level. Given an input instance $\boldsymbol{x}$, similar to Neural Cleanse, we define a generic trigger-injection function $B(\cdot)$:

$$B\left(\boldsymbol{x}, \boldsymbol{\lambda}', \Delta'\right) = \left(Z - \boldsymbol{\lambda}'\right) \odot \boldsymbol{x} + \boldsymbol{\lambda}' \odot \Delta' \tag{5}$$

where Z is an all-one matrix with the same size as the input instance $\boldsymbol{x}$. Δ' and $\boldsymbol{\lambda}'$ are both 3D matrices with the same size as the input instance $\boldsymbol{x}$ to be optimized. Values in the above matrices range from 0 to 1. And $\odot$ denotes the Hadamard product. Then, for every potential target class, we perform reverse engineering. Specifically, given an object class t to be analyzed and a potential target class tar, we obtain target label through Equation (6),

$$A(\boldsymbol{x}, t, tar)_{i,j} = \begin{cases} tar & if\ P(i,j) = t \\ P(i,j) & others \end{cases} \tag{6}$$

where t and tar are not identical. After obtaining the target labels of all instances, we compute a synthetic trigger $\left(\Delta', \boldsymbol{\lambda}'\right)$ that maximizes the proximity of predictions to target labels for all instances. Compared to Neural Cleanse, we reduce the number of synthetic triggers to $(K-1)^2$ when all object classes have been completely analyzed. This significantly reduces the amount of computation required in the reverse engineering stage. The final optimization formula is as follows:

$$\min_{\Delta', \boldsymbol{\lambda}'} loss\left(\mathcal{F}_\theta\left(B\left(\boldsymbol{x}, \boldsymbol{\lambda}', \Delta'\right)\right), A(\boldsymbol{x}, t, tar)\right) + \alpha \cdot ||\boldsymbol{\lambda}'||_F \tag{7}$$

where $loss(\cdot)$ is a loss function to measure the distance from predictions to target labels, such as cross-entropy. α is a weight factor mainly used to control the maximum degree of t segmenting to tar. $||\cdot||_F$ denotes frobenius norm.

4.3.2. Anomaly detection

To analyze a normal object class (except for background), we reverse engineer each possible target class (including background). Then, we perform anomaly detection for all normal object classes. In a semantic segmentation dataset, the background always consists of some non-goal normal classes. So the background is always scattered across the object class distribution space, which is different from other normal classes clustered in a location. That is why we can obtain a much smaller synthetic trigger when the target class is background than when a normal class becomes the target class during the experiments (we stipulate that $||\boldsymbol{\lambda}'||_F$ denotes the size of the synthetic trigger in this paper). We illustrate an abstract version of the cause resulting in the above phenomenon in Fig. 5. Therefore, as Fig. 10(a) in Section 5.4 reveals, the benign model will also be detected as an infected model if directly adopting the default setting in Neural Cleanse. For this reason, we adopt

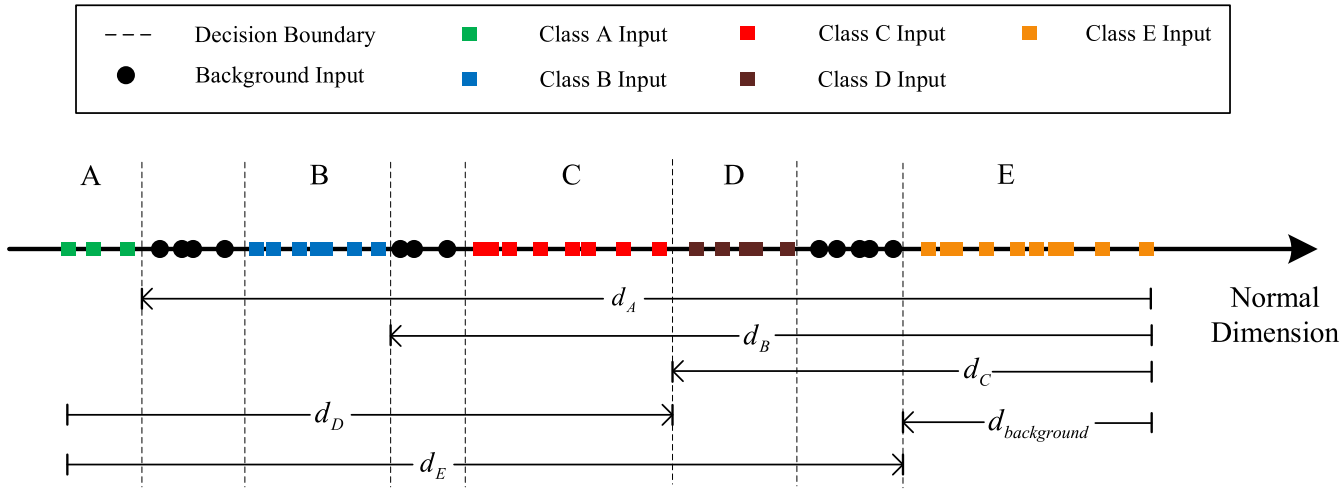

Fig. 5. A simplified illustration explaining that the synthetic trigger for the background is much smaller than that for the normal class. It shows the positions of objects in the input space and decision boundaries of the model for a 1-dimensional segmentation problem with six object classes (normal classes for triangles and the background for circles). d_i denotes the minimum modification needed to segment all normal objects into the target class i, which corresponds to the size of the synthetic trigger in reverse engineering. It is obvious that the background distribution scattering causes $d_{background}$ to be always smaller than others.

the idea in bootstrap aggregating (Breiman, 1996), weighted summation of standard anomaly index ($Index_{standard}$) and anomaly index ($Index_{normal}$) obtained by anomaly detection only for normal classes. Then, we mark the potential target class with an anomaly index greater than a predefined threshold as the target class, and the model is identified as infected. The corresponding generating synthetic trigger is considered the potential trigger. Assuming that our analysis is on "Class i" and "Class0" is the background, we can obtain $K-1$ synthetic triggers, *i.e.*, $\boldsymbol{\lambda}'_0,...,\boldsymbol{\lambda}'_{i-1}, \boldsymbol{\lambda}'_{i+1},...,\boldsymbol{\lambda}'_{K-1}$. The anomaly index of Segment Cleanse ($Index$) can be as follows:

$$Index = \omega_1 \times Index_{standard} + \omega_2 \times Index_{normal} \tag{8}$$

$$Index_{standard} = median\{\|\boldsymbol{\lambda}'_j\| - median\{\|\boldsymbol{\lambda}'_j\|\}_{j\neq i}\}_{j\neq i} \tag{9}$$

$$Index_{normal} = median\{\|\boldsymbol{\lambda}'_j\| - median\{\|\boldsymbol{\lambda}'_j\|\}_{j\neq i,0}\}_{j\neq i,0} \tag{10}$$

where $\{\cdot\}$ denotes a numerical series and $median(\cdot)$ can obtain the median of a numerical series. ω_1 and ω_2 are weighting factors in bootstrap aggregating. After extensive experiments, we find that when the weighting factor ratio is set to $9:1$ (*i.e.*, $\omega_1 = 0.9$, $\omega_2 = 0.1$), $Index > 2$ can be used as a condition to determine whether it is anomalous. Finally, we repeat the above operation for each normal class to detect the unknown model.

4.3.3. Model mitigation

In this stage, based on the previously obtained target classes and potential triggers, we detoxify the model to mitigate the impact of OFBA. The backdoor behavior on semantic segmentation can be regarded as a specific behavior displayed by the decoder when particular features appear. Since the encoder is only used to extract features, it is not directly involved in the attack from the perspective of OFBA. To this end, we freeze the encoder (*i.e.*, backbone) and detoxify only the decoder. This helps to alleviate the overfitting during detoxifying due to a large number of parameters and a limited number of examples. Specifically, we retrain the decoder with patched images, which are injected potential triggers and assigned with their ground truths.

5. Experiments

In this section, we first introduce the experiments setup including datasets, model structures, baseline selection, attack setup, and defense setup (Section 5.1). Then, we provide evaluation metrics for backdoor attacks and defenses on semantic segmentation (Section 5.2). Finally, we present and analyze the experimental results of OFBA and Segment Cleanse on three SOTA models (Section 5.3, 5.4).

5.1. Experiments setup

5.1.1. Datasets

We evaluate our attack and defense on the PASCAL VOC dataset (Everingham et al., 2010), the most widely used semantic segmentation benchmark. The PASCAL VOC dataset consists of internet images labeled with 21 object classes, including 20 normal object classes and the background. Following the division of the original dataset, we used 8498 examples for training and 2857 examples for testing. The image sizes are not uniform, so we adopt random clipping to adjust the size to 480×480. In addition, we also evaluate the performance of our attack on the Cityscapes (Cordts et al., 2016) dataset, which consists of road scenes captured from car-mounted cameras. It includes 19 normal object classes and the background (uniformly setting a small number of object classes as background). The testing set has 500 examples, and the training set totals about 23,000 examples. This dataset provides high-resolution images ($2,048 \times 1,024$ pixels), which require too much memory for some models. Thus, we resize all images to $1,024 \times 512$ and randomly clip the size to 480×480 during the training, then segment images with original sizes during the inference.

5.1.2. Model structures

We evaluate the proposed attack method on three state-of-the-art models, including the DeepLabv3 (Chen et al., 2017a), DenseASPP (Yang et al., 2018), and DaNet (Fu et al., 2019), respectively. We describe below the details of the above models and show the backbones we select in Table 2.

DeepLabv3 adopts dilated convolution and global pooling to improve segmentation accuracy and efficiency. It also incorporates an

Table 2
The details of semantic segmentation models used in this section.

Model	Backbone
DeepLabv3 (Chen et al., 2017a)	ResNet-50 (He et al., 2016a)
DenseASPP (Yang et al., 2018)	DenseNet-121 (Huang et al., 2017)
DaNet (Fu et al., 2019)	ResNet-50

ASPP component to perform convolutional operations at different scales to better handle objects of different sizes.

DenseASPP consists of a base network and a series of cascading convolutional layers. It combines the advantages of parallel and cascaded use of dilated convolution layers to produce more scale features at a larger scale.

DaNet is a dual attention network to capture global feature dependencies in the spatial and channel dimensions. It designs a positional attention component to learn the spatial interdependencies of features and a channel attention component to model channel interdependencies, resulting in significantly improved segmentation accuracy.

5.1.3. Baseline selection

For attack experiments, we selected a previous object-level attack, *i.e.*, FGBA (Li et al., 2021a) as the first baseline for attack experiments and used the same dataset and model structure for comparison. Considering the fixed immutable attacked class in FGBA during the inference, we set the attacked class uniformly to "person" for both attacks. We also provide the benign models that are no attack as the second baseline for attack experiments.

For defense experiments, to our knowledge, our work is currently the first backdoor defense strategy on semantic segmentation, and the vast majority of defense strategies in other fields, such as image classification (Gao et al., 2019; Qiao et al., 2019; Wu and Wang, 2021), cannot be applied to the semantic segmentation models. Therefore, we design our baselines to compare with our method. During the anomaly detection, to verify the validity of the proposed anomaly detection, we chose to use the results of the standard anomaly detection based on MAD directly as the first baseline for defense experiments. During the model mitigation, we choose the attack effect before retraining as the second baseline for defense experiments. In addition, models retrained with a small number of clean images (1,000 clean images forming the training set in our experiments) as the third baseline to compare with models defended by Segmentation Cleanse.

5.1.4. Attack setup

All our experiments are conducted on four NVIDIA RTX 2080 GPUs. On evaluating the effectiveness of backdoor attacks, considering the difference between background and normal object classes, we set the background and another object class (*e.g.*, æcowg in PASCAL VOC and æroadg in Cityscapes) as the target class, respectively. In addition, we set the poisoning rate $\gamma = 10\%$ for all experiments. The trigger is a black and white chessboard with size $3 \times 29 \times 29$, and the visible coefficient λ is set to 0.4. For OFBA, we inject the trigger into the cropped or resized clean images following Equation (3) and modify their labels following Equation (4). For FGBA, We inject the trigger into the bottom right corner of images and modify the class "person" to the target class in their labels. Moreover, during the training, we utilize the SGD optimizer, and the initial learning rate is set to 0.001 with a learning rate decay of 0.1 after every 20 epochs for all attack experiments. The batch size and training epochs are set to 4 and 100, respectively.

5.1.5. Defense setup

We evaluate the defense against OFBA. In practical application scenarios, we often need to protect only a few critical object classes (*e.g.*, persons encountered during automated driving) from the attack to reduce the number of synthetic triggers further. In our experiment, we choose the "person" as our protected object. Specifically, we randomly select 1000 clean images from the training set containing the "person". During the reverse engineer triggers, we set the weight factor α to 0.0001 and adopt the Adam optimizer to train 100 epochs at a constant learning rate of 0.005. We take an early stop to accelerate the defense speed when $||\boldsymbol{\lambda}'||_F$ no longer decreases. In the anomaly detection stage, we integrate anomaly indices by weighting coefficients of 0.9 and 0.1 and setting the threshold value to 2. Finally, in the model mitigation stage, we adopt the SGD optimizer, and the learning rate is set to 0.0001 to retrain. The batch size and training epochs are set to 4 and 20, respectively.

5.2. Evaluation metric

Semantic segmentation models classify each pixel in an image. In other words, its prediction is a collection consisting of many different object classes, which is quite different from classification. To fully understand the effectiveness of the proposed attack and defense, we evaluate the following metrics. The evaluation metrics consist of three parts: attack success rate, attacked test accuracy, and standard test accuracy, which are used to evaluate what is expressed in Equation (1), (2).

- Attack success rate (ASR) is the most decisive metric for evaluating the success of a backdoor attack. As shown in Equation (11),(12), we adopt PA_t and IoU_t to represent it. In general, an effective attack should have a high ASR, and a useful defense ought to hold a low ASR. Note that IoU_t is a relatively special metric. A high IoU_t is insufficient to mean a successful attack when the result no longer has a high PA_t. For example, images from the PASCAL VOC dataset have a large area covered by background, and IoU_t will be high even if the attack is ineffective or the defense is effective when the background is the target class.
- Attacked test accuracy (ATA) evaluates the segmentation performance of benign objects (*i.e.*, normal object classes excepting for attacked classes in poisoned images). Equation (13),(14) shows two indicators in ATA, namely mPA_p and $mIoU_p$. A reasonable attack or defense, based on its ability to attack an object or defend against a backdoor attack, should not disproportionately affect the performance of other benign objects.
- Standard test accuracy (STA) is the accuracy of the test data. It compose of mPA_b and $mIoU_b$, which are display in Equation (15),(16). The STA of an infected or defended model ought to be similar to the test accuracy of the benign model.

$$PA_t = \frac{n_{t,tar}}{n_t} \tag{11}$$

$$IoU_t = \frac{N_{tar,tar}}{\sum_{i \neq tar} N_{i,tar} + \sum_{i \neq tar} N_{tar,i} + N_{tar,tar}} \tag{12}$$

$$mPA_p = \frac{1}{k-2} \sum_{i \neq t,tar} \frac{N_{i,i}}{N_i} \tag{13}$$

$$mIoU_p = \frac{1}{k-2} \sum_{i \neq t,tar} \frac{N_{i,i}}{\sum_{j \neq i} N_{j,i} + \sum_{j \neq i} N_{i,j} + N_{i,i}} \tag{14}$$

Table 3
The effectiveness (%) of our attack against segmentation models on the PASCAL VOC dataset and Cityscapes dataset.

Target Class	Model	Metric					
		PA_t	IoU_t	mPA_p	$mIoU_p$	mPA_b	$mIoU_b$
Background (Class 0)	DeepLabv3	98.72	92.77	52.84	27.85	91.45	70.77
	DenseASPP	97.32	92.58	62.87	31.35	90.45	67.11
	DANet	98.85	93.83	68.95	36.24	91.30	70.90
Cow (Class 10)	DeepLabv3	86.19	72.58	87.78	50.92	91.65	71.38
	DenseASPP	81.18	69.20	85.26	43.46	90.61	67.65
	DANet	85.36	74.90	89.08	50.66	91.22	69.62

Table 4
The effectiveness (%) of our attack against segmentation models on the PASCAL VOC dataset and Cityscapes dataset.

Target Class	Model	Metric					
		PA_t	IoU_t	mPA_p	$mIoU_p$	mPA_b	$mIoU_b$
Background (Class 0)	DeepLabv3	82.15	72.67	85.47	54.36	86.09	56.74
	DenseASPP	74.53	70.57	84.21	54.48	86.22	58.64
	DANet	84.00	70.71	84.32	51.14	85.57	53.73
Road (Class 0)	DeepLabv3	82.52	64.94	83.62	52.67	85.62	55.97
	DenseASPP	82.41	65.62	83.36	51.07	86.32	53.88
	DANet	81.49	77.43	82.37	49.18	85.32	53.67

$$mPA_b = \frac{1}{k}\sum_i \frac{n_{i,i}}{n_i} \tag{15}$$

$$mIoU_b = \frac{1}{k}\sum_i \frac{n_{i,i}}{\sum_{j\neq i} n_{j,i} + \sum_{j\neq i} n_{i,j} + n_{i,i}} \tag{16}$$

where t and tar denote the attacked and the target classes, respectively. We assume that there exists a k object class in the image. $n_{i,j}$ and $N_{i,j}$ denote the number of pixels where i is predicted to be j in the ground truth and target label, respectively. n_i and N_i denote the number of pixels belonging to Class i in the ground truth and target label, respectively.

5.3. Main results for attacks

5.3.1. Attack effectiveness

On evaluating OFBA, we experimented with randomly selecting the attacked class for each image during the inference. As shown in Table 3, 4, our attack can successfully create backdoors with a high ASR on all semantic segmentation models by poisoning only a small proportion (10%) of training data. Meanwhile, it maintains high performance in predicting benign objects and clean images. Specifically, our attack can achieve a $PA_t > 74\%$ and an $IoU_t > 64\%$ on both datasets. Besides, it still holds over 85% on mPA_b and 53% on $mIoU_b$ on clean images. In addition, comparing different target classes, we find the model is much easier to attack when the background is the target class than when a normal object class is the target class. This also seems to suggest the conclusion of Fig. 5 in Section 4.3. Moreover, in terms of the ATA metric (*i.e.,* mPA_p and $mIoU_p$), our method does not perform very well on the PASCAL VOC dataset. It means that when triggers are injected into several objects, the accuracy of otherwise benign object classes (*i.e.,* mPA_p and $mIoU_p$) is reduced by 16.65% (average) on mPA and 29.49% (average) on $mIoU$ compared to images without trigger injection (*i.e.,* mPA_b and $mIoU_b$). This may be because the trigger is often challenging to inject into fine-size objects in our attack. Most images in the PASCAL VOC dataset consist of a single main object as the subject combined with other fine-size objects. It will result in most of the remaining benign objects being fine-size objects in effective poisoned images (*i.e.*, images have successfully injected the trigger). Normally, the segmentation performance for fine-size objects tends to be poor because their contours are too small to be accurately positioned.

To measure the difference between the different attacked classes, we attack the same object class for each image during the inference. Fig. 6 shows the PA_t of different attacked classes. Clearly, the performance of our backdoor attack yields significant differences across attacked classes. Especially on the Cityscapes dataset, there even exist object classes that OFBA fails to attack. This is because the sizes of some objects are so small that it is difficult to inject the trigger into images. An interesting phenomenon is that, on different model structures, the distributions of attack success rates on different attacked classes are incredibly similar. It illustrates that the backdoor attack is a dataset-specific response unrelated to the models, consistent with the classification domain (Gu et al., 2019; Li et al., 2021b).

5.3.2. Comparison with baselines

We compare the proposed attack method with baselines for attack experiments described in Section 5.1.3. The performance comparison between the baselines and our attack is presented in Table 5. As can be observed, in comparison with FGBA, our method accomplishes a more malicious backdoor attack at a cost of only 2.72% (average) on PA_t, 6.92% (average) on IoU_t, 13.30% (average) on mPA_p, and 11.04% (average) on $mIoU_p$ reduction. Meanwhile, for the STA metric, we improve by 0.07% (average) on mPA_b and 1.68% (average) on $mIoU_b$, which means that our attack method is less destructive to model accuracy when creating a backdoor than previous work. Besides, on the other baseline, the ASR metric on benign models (*i.e.,* no attack) is only 3.72% (average) on PA_t, and the STA metric is 91.63% (average) on mPA_b and 68.23% (average) on $mIoU_b$. Clearly, the benign models can segment poisoned images correctly; that is to say, the trigger affects the model and is not destructive to the image itself.

5.3.3. Effects of different structures

The structure of a semantic segmentation model is usually composed of multiple different components stacked on each other. The impact of these components on OFBA has yet to be discovered. In

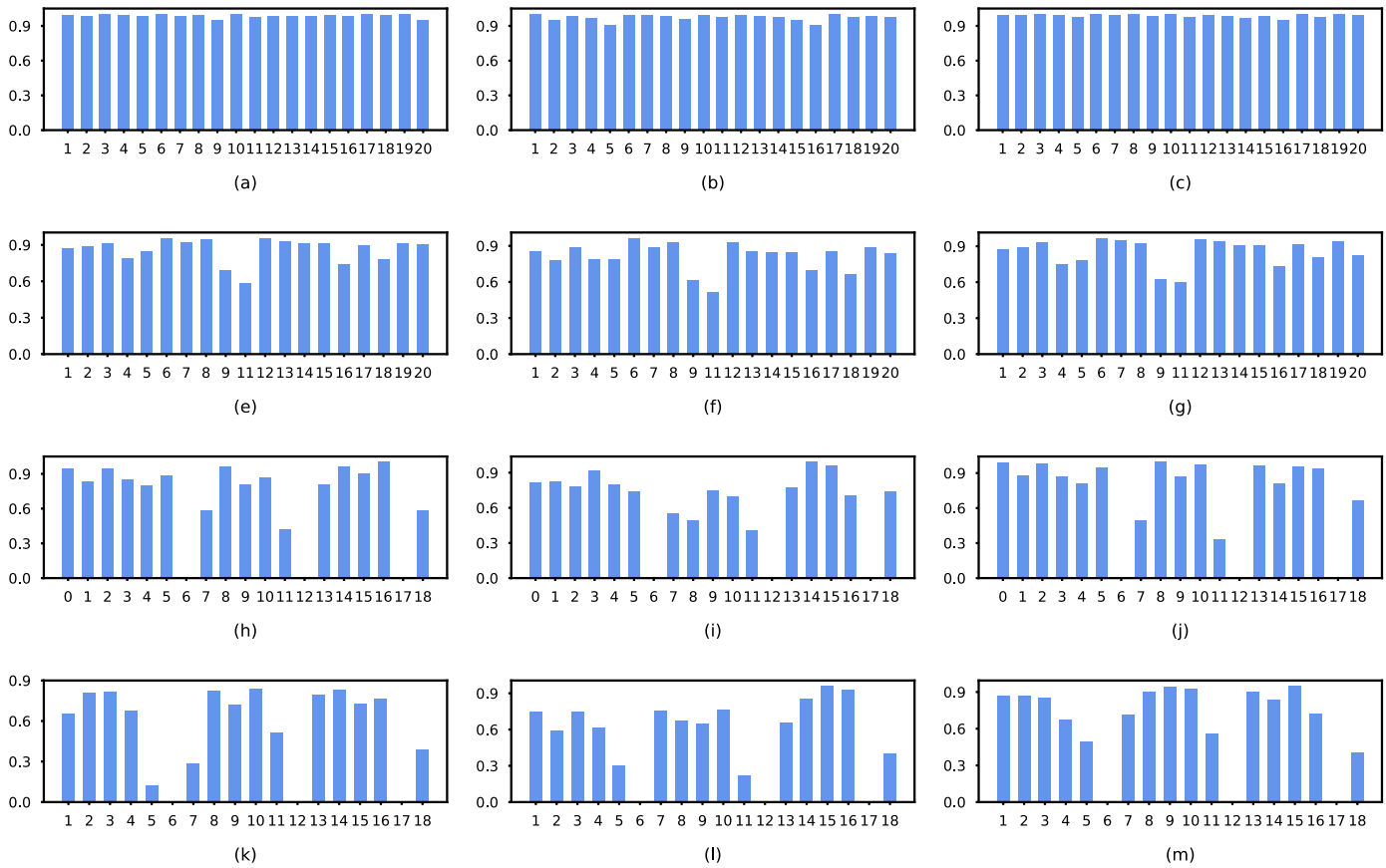

Fig. 6. The PA_t of different models with different attacked classes. (a)-(c): Models with targeted background on the PASCAL VOC dataset. (d)-(g): Models with targeted "cow" on the PASCAL VOC dataset. (h)-(j): Models with targeted background on the Cityscapes dataset. (k)-(m): Models with targeted "road" on the Cityscapes dataset.

Table 5
The comparison of different methods against the "person" class on the PASCAL VOC dataset with targeted background and "cow" class, respectively.

Target Class	Model	Attack Method	Metric					
			PA_t	IoU_t	mPA_p	$mIoU_p$	mPA_b	$mIoU_b$
Background (Class 0)	DeepLabv3	no attack	9.62	65.51	83.42	53.20	92.52	70.83
		FGBA (Li et al., 2021a)	98.84	95.95	75.22	47.18	90.87	64.28
		ours	99.00	91.54	71.41	38.09	91.46	70.13
	DenseASPP	no attack	6.75	58.72	79.83	49.52	90.44	65.15
		FGBA	98.12	94.34	72.91	36.92	90.02	63.34
		ours	95.26	89.66	50.07	29.22	90.17	64.37
	DANet	no attack	3.86	61.05	82.11	51.80	91.97	68.72
		FGBA	98.51	96.34	85.62	53.08	91.72	67.91
		ours	98.71	92.09	72.13	33.07	91.20	68.54
Cow (Class 10)	DeepLabv3	no attack	0.90	0.89	92.60	71.02	92.52	70.83
		FGBA	92.41	86.21	81.33	48.82	92.04	69.35
		ours	91.50	79.41	78.79	39.81	92.49	69.96
	DenseASPP	no attack	1.16	1.01	90.82	65.16	90.42	65.11
		FGBA	93.25	84.88	79.53	41.61	90.38	64.14
		ours	84.80	70.74	45.30	26.55	90.28	64.26
	DANet	no attack	0.02	0.89	91.96	68.80	91.91	68.72
		FGBA	94.98	85.87	81.08	50.81	91.22	66.38
		ours	90.52	78.62	78.17	45.43	91.04	68.19

this section, we show observations about whether residual structures, skip-connections, and dilated convolutions affect OFBA on the PASCAL VOC dataset.

We consider the simplest FCN32s (Long et al., 2015) as the base structure to which other related components will be added or modified. In addition, to exclude the interference caused by the increase in model accuracy after adding components, which can affect the results of OFBA, we use the predictions of clean images instead of ground truths as a reference to evaluate poisoned images when evaluating the ASR metric. Fig. 7, 8, 9 show the variation of ASRs on different visible coefficients after adopting residual structures, skip-connections, and dilated convolutions, respectively. The higher the ASR is, the more vulnerable the model is. It is evident that these structures always make the model more vulnerable

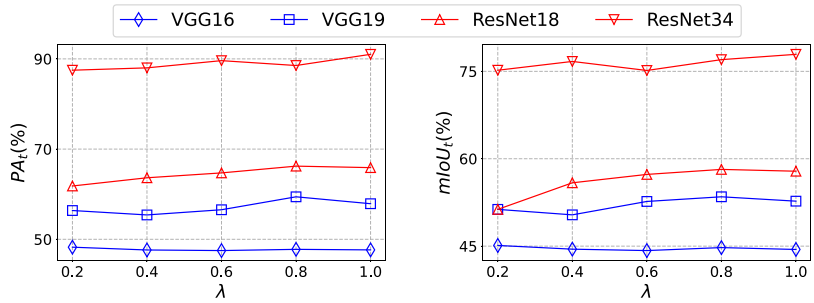

Fig. 7. The effect of residual structure towards OFBA with different visible coefficients. The four models in the above figure are variations on their backbones based on FCN8s. The red lines denote ResNet networks with the residual structure. The blue lines denote VGG networks, one of the most classical convolutional neural network structures. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

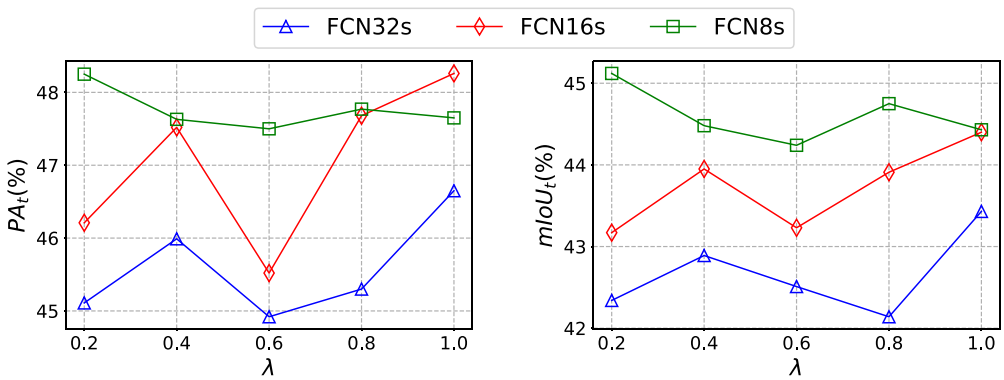

Fig. 8. The effect of skip-connection towards OFBA with different visible coefficients. We use VGG16 as the backbone for all models and FCN32s as the control group in this experiment. FCN16s fuses the information of the bottom feature map through a skip-connection, while FCN8s adopts two skip-connections.

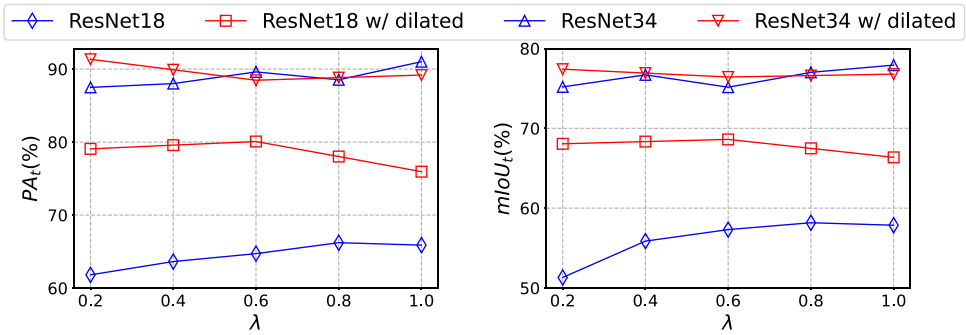

Fig. 9. The effect of dilated convolution towards OFBA with different visible coefficients. The four models in the above figure are all based on FCN8s. The red lines denote that the dilated convolutions are used in network structures and the blue lines are the control groups in this experiment. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to OFBA while improving the accuracy of models. It is worth noting that, in evaluating residual structure, the parameter quantities of four backbones are ordered from smallest to largest as VGG16 (Simonyan and Zisserman, 2015), ResNet18 (Ye et al., 2022), VGG19, and ResNet34. In general, the larger the number of parameters, the better the model recognizes small features (triggers always have small sizes), and the more vulnerable the model is to attack. It can be seen that the parameter quantity will have some influence on the experimental results, but its impact will not overturn the above conclusions. On the contrary, we find that the effect of residual structures is enormous compared to that of the parameter quantity towards OFBA.

5.4. Main results for defense

5.4.1. Performance of anomaly detection.

To confirm the value and effectiveness of our work, we calculate standard anomaly indices (*i.e.*, the first baseline for defense experiments mentioned in Section 5.1.3) and our proposed anomaly indices for six infected models and their matching benign models after "Reverse Engineer Triggers". Recall that our anomaly index threshold for infection is set to 2. As shown in Fig. 10a, standard anomaly indices of benign models, except for deeplabv3, are greater than 2, which means the detection incorrectly marks them as infected. And the phenomenon confirms our analysis in

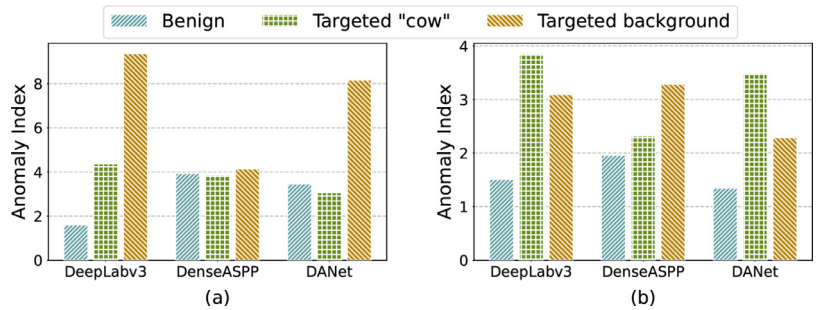

Fig. 10. The anomaly indices of benign and infected models. The benign models (blue columns) are trained with original datasets, and the infected models are attacked by OFBA with "cow" (green columns) and background (yellow columns) as the target class, respectively. (a): Standard anomaly indices based on MAD; (b): our anomaly indices based on bootstrap aggregating. The smaller the index, the harder the attack to detect. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 6
The comparison of models before and after retraining with clean images or patched images on the PASCAL VOC dataset.

Target Class	Model	Training Exsamples	Metric					
			PA_t	IoU_t	mPA_p	$mIoU_p$	mPA_b	$mIoU_b$
Background (Class 0)	DeepLabv3	no training	99.0	91.5	71.4	38.1	91.5	70.1
		clean examples	90.9	92.7	60.7	27.9	89.2	65.4
		patched examples	70.5	84.9	68.0	38.0	88.7	63.8
	DenseASPP	no training	95.3	89.7	50.1	29.2	90.2	64.4
		clean examples	88.4	92.2	61.7	26.1	88.9	63.2
		patched examples	64.3	85.9	70.8	39.2	87.1	62.6
	DANet	no training	98.7	92.1	72.1	33.1	91.2	68.5
		clean examples	96.2	92.9	61.1	35.3	88.5	62.5
		patched examples	26.0	69.9	83.3	44.1	87.6	61.6
Cow (Class 10)	DeepLabv3	no training	91.5	79.4	78.8	39.8	91.5	70.0
		clean examples	88.5	81.5	58.8	39.3	90.3	65.4
		patched examples	20.9	21.9	82.8	51.0	88.9	63.8
	DenseASPP	no training	84.8	70.7	45.3	26.6	90.3	64.3
		clean examples	79.8	71.0	59.2	30.7	88.0	63.2
		patched examples	23.2	29.9	82.0	46.2	87.5	62.6
	DANet	no training	90.5	78.6	78.2	45.4	91.0	68.2
		clean examples	90.5	77.4	57.0	37.5	88.3	62.5
		patched examples	21.8	28.8	81.8	47.2	86.7	61.6

Section 4.3.2. Fig. 10b shows the results of anomaly detection using the proposed anomaly index defined as Eq. 8. We keep the indices in detecting benign models below the threshold, that is to say, our detection does not make the mistakes mentioned above. Meanwhile, all infected models can be detected on account of their indices above the threshold. With all being discussed above, our detection overcomes the negative effects of the peculiarities of the background distribution on detection.

5.4.2. Performance of model mitigation

In this section, we experimentally demonstrate that our method can make infected models somewhat resistant to the backdoor attack. As shown in Table 6, comparing with models no retraining (*i.e.*, the second baseline for defense experiments described in Section 5.1.3), we significantly reduce values on ASR in all models without sacrificing accuracy on STA. In particular, against infected models with targeted a normal object class (*e.g.*, "cow"), we reduce ASR to 21.97% (average) on PA_t and 26.87% (average) on IoU_t. Besides, the reduction of STA is only 3.2% (average) on PA_b and 4.92% (average) on $mIOU_b$. An interesting phenomenon is that the defended models outperform the non-defended models in terms of ATA metrics, which improves mPA_p by 12.13% and $mIoU_p$ by 8.92%, respectively. As mentioned in Section 5.3.1, the attack reduces the performance of models on the ATA metric, and the above phenomenon also implies the defended models moving towards benign models. On the third baseline for defense experiments, we find that models retrained with clean examples could somewhat mitigate OFBA. This is because, after retraining, the decoder mitigates or even forgets the sensitivity response to the trigger. However, models retrained with patched images (*i.e.*, defended models) can defend OFBA more significantly, which reduces by 51.27% on PA_t and 31.07% on IoU_t. This also shows that the patched examples we have produced are better-retrained examples than clean examples. Besides, it is always easier to detoxify an infected model where a normal class is the target class than another infected model where the background is the target class. It is consistent with the conclusion in Section 5.3.1 that the background is a better target class from the attacker's perspective.

6. Conclusion

In this work, we report limitations of existing backdoor attacks on semantic segmentation, which either can only operate on predictions from the image level or can only attack with a pre-defined fixed attacked class. We explore a new backdoor attack that can maliciously manipulate its predictions by selecting the attacked class during the inference. Experiments demonstrate that the approach performs effectively and more maliciously on existing semantic segmentation models. In addition, we analyze the impact of three common semantic segmentation structures on backdoor robustness and reveal that the higher the benign accuracy, the more vulnerable it is to attack as a general rule. Moreover, to address the gap of backdoor defense on semantic segmentation, we summarize why trigger synthesis methods are not applicable, for which we propose Segment Cleanse to defend against OFBA. Experiments demonstrate that our defense is reliable and effective. In the

future, our work will force on the following areas: (1) more stealthy backdoor attacks for the human eyes on semantic segmentation, (2) exploring and summarizing the impact of more structures on backdoor robustness, and (3) more effective and stable defense against the backdoor attack on semantic segmentation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jiaoze Mao: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Yaguan Qian:** Conceptualization, Methodology, Resources, Data curation, Writing – review & editing, Funding acquisition, Project administration. **Jianchang Huang:** Validation, Investigation, Writing – review & editing. **Zejie Lian:** Validation, Investigation, Formal analysis. **Renhui Tao:** Validation, Investigation, Writing – review & editing. **Bin Wang:** Conceptualization, Supervision, Project administration, Funding acquisition. **Wei Wang:** Conceptualization, Supervision, Project administration. **Tengteng Yao:** Validation, Investigation.

Data availability

No data was used for the research described in the article.

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